

IRVIN SIVYA

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EDUCATION

University of Waterloo

Bachelor of Computer Science (Co-op)

Sept 2026 – 2031

Waterloo, ON

- Incoming Computer Science co-op student; recipient of **\$20,000 in entrance scholarships**.

Rick Hansen Secondary School

Ontario Secondary School Diploma

2022 – 2026

Mississauga, ON

- **Highest graduating average (~99%) in a class of ~460**; Schulich Leader Scholarship nominee (**top STEM student**); AP Scholar with 5s in Calculus AB, Calculus BC, and Computer Science A.

EXPERIENCE

Software Engineering Intern | Python / Flask / MongoDB / Chart.js

Summer 2025

SalesPatriot (YC W25), via Harvard Undergraduate Ventures-TECH (HUVTSP)

Remote

- Built a **full-stack evaluation dashboard** with a cohort partner to flag, categorize, and track hallucinations across the company's large language model (LLM) pipeline, turning unstructured model outputs into queryable records and giving the team ongoing visibility into model reliability.
- Presented the tool and its findings to the founder, translating technical results into concise, business-focused recommendations; completed Harvard's HUVTSP entrepreneurship curriculum.

Lead Programmer & Mentor | Java / WPILib / Phoenix 6 / CTRE / AdvantageScope

2022 – Present

FIRST Robotics Competition (FRC) Teams 1285 & 1241

Mississauga, ON

- Lead programmer for **4 years (2,000+ hours)** on 115-lb competition robots; implemented PID control, motion profiling, odometry, computer vision tracking, and path following as **real-time decision loops running every 20 ms**.
- Instrumented robots with structured telemetry logging and analyzed match logs in AdvantageScope to diagnose control failures and tune parameters between matches, turning sensor time-series into design changes.
- Won the **FRC Dean's List Award at Districts (2024)**; contributed to 15+ team awards and **competed internationally 3x**.
- Write technical documentation and mentor programmers on Team 1241, breaking down control theory and vision concepts into teachable explanations for newcomers; Workshops and Games Lead for the 2025 Girls+ in STEAM Olympics (GiSO).

Robotics, Coding & STEM Instructor (Part-Time)

2025 – Present

Code Ninjas / City of Brampton

Brampton, ON

- Deliver hands-on instruction in programming, robotics, and math to ages 5–14; coached 4 FIRST LEGO League teams (40 students) and serve as a fill-in instructor across the City of Brampton STEM catalogue.

PROJECTS

CrimeWatcher | Python / Pandas / Scikit-learn / Matplotlib / PostgreSQL (Supabase) / Next.js / Leaflet / Expo

- Built an **end-to-end machine learning pipeline** for predictive modeling of crime risk across 158 Toronto neighbourhoods from a **474K-record Toronto Police dataset**, engineering time-series lag, seasonal, and spatial features in Pandas.
- Trained and tuned a gradient-boosted regression model, **testing on the final 6 months of data to prevent leakage** in the time series.
- Deployed the model behind a live Next.js/Leaflet dashboard backed by PostgreSQL (Supabase), turning raw incident records into neighbourhood-level insights; a companion Expo app streams real-time distress signals onto the map.

Project R.O.S.E. (1st Place, WolfHacks 2025) | Python / Hugging Face Transformers / Gradio / Java / Micro:bit

- Won **Peel's largest hackathon (76 teams)** with a 3-part waste-management solution: an image-classification model for waste sorting served through Gradio, a Java awareness app, and a Micro:bit prototype of a plastic-degrading bacteria machine.

ExpenseCity (2nd Place, IBM Bobathon) | IBM Bob / IBM Watsonx / JavaScript / Three.js

- Extended a three-part expense tracker with a 3D visualization that encodes expense size as building height, making spending patterns readable at a glance.
- Built with a small team during a fast-paced **3-hackathon sprint over 5 days**, placing in all three events (2nd, 2nd, 1st).

SKILLS, CERTIFICATIONS & AWARDS

- **Data & Analytics:** Python, Pandas, Scikit-learn, Matplotlib, SQL, feature engineering, time-series modeling, gradient-boosted trees, train/test design, model evaluation, data cleaning.
- **Core Concepts:** Data structures and algorithms, data modeling, predictive modeling, data analysis and insight generation, software development lifecycle (SDLC), model deployment.
- **Machine Learning & LLMs:** Hugging Face Transformers, TensorFlow, Keras, OpenCV, Gradio, IBM Watsonx, RAG and vector search, fine-tuning (LoRA), prompt engineering, hallucination testing.
- **Languages:** Python, Java, SQL, TypeScript, JavaScript, HTML/CSS.
- **Data Stores, Backend & Tools:** PostgreSQL, Supabase, MongoDB, Flask, Node.js, Next.js, React, Expo, Git/GitHub, Vercel.
- **Certifications:** AWS Certified AI Practitioner (AIF-C01) and AWS Certified Cloud Practitioner (CLF-C02), both July 2026.
- **Awards:** Highest graduating average, 2025 CSMC School Champion, 4x hackathon winner, FRC Dean's List Award, Ontario Principals' Council Award.